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Boosting and foundation-model adaptation

Week 11: Sequential ensembles, then the full adaptation ladder from zero-shot to LoRA.

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By the end of this week you should be able to

Learning outcomes served: LO1, LO2, LO3, LO5

- Explain boosting as sequential error correction and contrast it with bagging.
- Describe gradient boosting as fitting successive models to the residuals of the current ensemble.
- Place zero-shot use, feature extraction, fine-tuning and parameter-efficient fine-tuning on a single ladder of increasing cost and data requirement.
- Compute the trainable-parameter count for a LoRA configuration and compare it with full fine-tuning.
- Choose an adaptation strategy given a labelled-data budget, a compute budget and an interpretability requirement.

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The biological problem

Part 1 · Why anyone cares about this week's method

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Biological context

- Gradient-boosted trees are, in practice, the strongest general method for tabular biological data, and they remain competitive with deep learning on that data type. Meanwhile, for sequence data, the practical question has become how to adapt a pretrained model to a task with a few hundred labels.

- This week covers both, because both are answers to the same question: how do you get good performance without a large labelled dataset?

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The method

Part 2 · Concepts, then the mathematics

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Boosting

Fit a weak learner, find what it got wrong, fit the next learner to concentrate on those cases, repeat. Bagging builds its models in parallel and independently; boosting builds them in sequence and dependently.

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Gradient boosting

Each new tree is fitted to the negative gradient of the loss with respect to the current predictions. For squared error that is simply the residuals. XGBoost, LightGBM and CatBoost are engineering refinements of this idea.

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Key concepts

Boosting overfits differently A random forest is difficult to overfit by adding trees. A boosted ensemble is not: too many stages, or too high a learning rate, will fit noise. Early stopping on a validation set is not optional.

Zero-shot Use the pretrained model with no task-specific training at all. Requires the task to be expressible in terms the model already handles. Costs no labels, and the ceiling can be low.

Feature extraction Freeze the model, take embeddings, train a small classical head. Works with a few hundred labels, needs no GPU after the embeddings are computed, and is the baseline everything else must beat.

Fine-tuning Continue training the model's weights on your task. Highest ceiling, highest cost, and on small biological datasets a real risk of catastrophic forgetting and of overfitting a very large parameter count to very few examples.

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Key concepts (continued)

Parameter-efficient fine-tuning Update a small number of parameters and freeze the rest. Adapters insert small trainable modules between layers. LoRA injects trainable low-rank matrices, leaving the pretrained weights untouched.

LoRA Instead of updating a weight matrix W of size d by k, learn A of size d by r and B of size r by k with r much smaller than both, and use W + BA.

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Gradient boosting

$$F_m(x) = F_{m-1}(x) + \nu \cdot h_m(x)$$

Each new learner is fitted to the negative gradient of the loss with respect to the current predictions. For squared error that is simply the residuals.

- F_m the ensemble after m stages
- h_m the new weak learner, usually a shallow tree
- ν the learning rate, typically 0.01 to 0.1; smaller needs more stages
- `contrast` bagging is parallel and independent; boosting is sequential and dependent, so it must be stopped early

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LoRA: low-rank adaptation

$$W' = W + BA$$

$$A \text{ is } d \times r, \quad B \text{ is } r \times k, \quad \text{with } r \ll \min(d, k)$$

Freeze the pretrained weight matrix W and learn only the low-rank update BA. On GPT-3 175B this reduced trainable parameters by a factor of 10,000 and GPU memory by 3, with no added inference latency because BA folds back into W after training.

W the frozen pretrained weight matrix, d × k

r the rank; 4 to 64 in practice

per matrix r(d + k) trainable parameters instead of d·k

adapters the alternative: small trainable modules inserted between frozen layers, which do add latency

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The LoRA parameter count, worked

$$12 \text{ layers} \times 2 \text{ matrices} \times 8 \times (768 + 768) = 294{,}912$$

A 12-layer model with hidden size 768, LoRA at rank 8 on the query and value projections. Against a 117-million-parameter backbone that is 0.25 per cent.

$r(d+k)$	$8 \times 1{,}536 = 12{,}288$ parameters per adapted matrix
$\times 2$	query and value projections per layer
$\times 12$	layers
result	294,912 trainable, 0.25% of the backbone

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The adaptation ladder

Climb one rung at a time, and only when the rung below has demonstrably run out.

Rung	Labels needed	Compute	The risk
Zero-shot	None	Inference only	The ceiling can be low, and you cannot tell in advance
Frozen embeddings + classical head	100 to 1,000	One forward pass, then a CPU	Almost none. Most projects should stop here
Parameter-efficient (LoRA, adapters)	1,000 to 10,000	A GPU for minutes to hours	New hyperparameters, each an opportunity to overfit
Full fine-tuning	10,000 and above	A GPU for hours to days	Fits a huge parameter count to little data; hard to reproduce

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Real measured learning curve, WDBC

Steep gains early, flat later. Mean of 20 stratified resamples at each size. That flattening is why the labelled-data budget, not the model, usually decides which rung you can reach.

Labelled examples	Logistic regression	Random forest
11	89.3%	88.4%
28	93.9%	92.6%
57	95.4%	94.3%
114	96.5%	95.3%
228	97.2%	95.7%
455	97.6%	96.1%

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Worked example

Part 3 · The adaptation ladder, with the arithmetic

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The adaptation ladder, with the arithmetic

- The rungs, in increasing order of cost:
- Zero-shot. No labels, no training. Try it first, because it is free and it occasionally suffices.
- Feature extraction. A few hundred labels, one forward pass per example, then a logistic regression. This is the rung most projects should stop at.
- Parameter-efficient fine-tuning. Low thousands of labels, a GPU for an hour. Take a model with 12 layers and hidden size 768, and apply LoRA at rank 8 to the query and value projections. Each adapted matrix contributes $r(d + k) = 8(768 + 768) = 12{,}288$ parameters.

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The adaptation ladder, with the arithmetic (continued)

- Two projections per layer across 12 layers gives 12 x 2 x 12,288 = 294,912 trainable parameters. Against a 117-million-parameter backbone that is roughly 0.25 per cent. The simulator on this page recomputes this exactly for whatever configuration you set.
- Full fine-tuning. Tens of thousands of labels, real GPU time, and every parameter in play. On a few hundred biological examples this is usually the wrong tool, because you are fitting a hundred million parameters to a few hundred points and the validation protocol has to be correspondingly strict.
- The general shape of the learning curve is not a matter of opinion: it was measured on real data for this module. On WDBC, logistic regression goes from 89.3 per cent accuracy with 11 labelled examples to 96.5 per cent with 114 and 97.6 per cent with 455. Steep gains early, flat later.
- That diminishing return is why the labelled-data budget, not the model, usually decides which rung you can reach.

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Climb the ladder one rung at a time, and only when the rung below has demonstrably run out.

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Try this interactively

Fine-Tuning Simulator

What does each rung of the adaptation ladder really cost?

`weeks/week11.html` · `Interactive exploration`

Open the week page on the course website and scroll to Interactive exploration.

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The foundation-model connection

Part 4 · A note on what the simulator is

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A note on what the simulator is

- Fine-tuning a real foundation model cannot happen in a browser tab. The simulator on this page therefore does two different things, and it is worth being clear about which is which.
- The parameter arithmetic is exact: given your architecture and rank, it computes the true trainable-parameter counts and memory ratios.
- The performance curves are a transparent parametric model of learning-curve behaviour, with the assumptions written on screen, calibrated so that its qualitative behaviour matches the real learning curve measured on WDBC.
- It is a model of how these methods behave, not a record of measured results, and it says so.

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Interpretation and practical

Part 5 · What the result means, and what you will do with it

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You have 340 labelled sequences and access to one GPU for a day. Argue for the adaptation strategy you would choose, then argue against it. Which argument is stronger, and what single piece of evidence would settle it?

Think like a scientist. Bring your answer to the laboratory.

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Boosting, and costing an adaptation strategy

This week's laboratory · Tune a boosted ensemble properly, then cost out the four rungs of the adaptation ladder for a realistic project. · 2 hours

- Fit GradientBoostingClassifier on WDBC. Sweep learning_rate and n_estimators over a grid and plot validation accuracy as a heatmap. Identify the trade-off between the two.

- Add early stopping with a validation fraction. Show a configuration that overfits without it and does not overfit with it.

- Compare gradient boosting with the random forest from Week 10 on accuracy, training time and sensitivity to hyperparameters. Which would you rather hand to a collaborator, and why?

- On the protein set, train logistic regression on the frozen embedding for the hasSite label. This is the feature-extraction rung. Record accuracy and wall-clock time.

- Using the LoRA arithmetic from the lecture, compute the trainable-parameter count for a 12-layer, 768-hidden model at ranks 4, 8, 16 and 64 applied to query and value projections. Express each as a percentage of a 117-million-parameter backbone.

- Write a one-page adaptation plan for a project with 340 labelled sequences and one GPU-day. State which rung you would use, what you would measure to know whether to climb higher, and what would make you stop.

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Self-check

- What is the essential difference between bagging and boosting?
- A) Boosting uses more trees
- B) Bagging trains models independently in parallel and averages them; boosting trains them sequentially, each correcting its predecessors
- C) Boosting is always more accurate
- D) Bagging cannot use trees

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Self-check

- LoRA at rank 8 on query and value projections, 12 layers, hidden size 768. How many trainable parameters?
- A) 12,288
- B) 294,912
- C) 1,179,648
- D) 117,000,000

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References and further reading

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Questions

Next week: continue to Week 12.

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